

Survey Assignment

Minerva University

NS51: Empirical Analyses

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Survey Assignment

Supporting Action of the Climate Action Plan and a Concrete Solution

To “build a fast and reliable transit system that will be everyone’s preferred way to get around,” our group focused on promoting electric bikes (e-bikes) as a sustainable transportation solution (*SAN FRANCISCO’S CLIMATE ACTION PLAN 2021*, 2015). San Francisco’s steep hills and rugged terrains, a constraint, make it challenging for some individuals to use regular bikes. Therefore, e-bikes offer a viable alternative that can tackle the geographical constraint of San Francisco (SF) while providing an eco-friendly mode of transportation.

In order to devise an effective solution, it is crucial to identify and evaluate the existing solutions that promote e-bikes. We approached this in two ways: 1) investigate how the California government has promoted e-bike usage, and 2) conduct a case study of foreign countries’ e-bike solutions that can be modified to implement in SF.

The California government introduced a program to offer free e-bikes to low-income residents from Berkeley and is planning to expand this program to California residents by allocating \$13M as a subsidy when buying a new e-bike (Kwok, 2023). In San Francisco, the city government is affiliated with Lyft, under the name “Bay Wheels.” The government is planning to construct more hybrid e-bike stations for better accessibility (San Francisco Municipal Transportation Agency, 2019). However, both California and SF governments are not providing direct subsidies for renting e-bikes.

Recognizing the lack of subsidy for e-bike renting fees, foreign countries’ exemplary subsidizing policy for bike rental services needs to be addressed to further discuss our solution’s practicality. In South Korea, Seoul’s city government provides \$0.3 for each e-bike rental fee to its citizens (Kim, 2011). Still, we have to consider that the price for a single e-bike rental fee in

Seoul (\$0.5 per 30 minutes) is significantly lower than that in SF (\$12.49 per 30 minutes), which requires appropriate adjustment specific to SF's situation (Seoul Bike, n.d.).

Based on this analysis of pre-existing solutions, we propose a subsidy program to reduce the rental fees for e-bikes provided by BayWheels. We recommend lowering the hourly rental fee from the current \$12 to a range of \$2.5 to \$3.5. We considered the average price of gasoline in the US (\$3.606 per gallon) and the single ride fare for Muni (\$2.50) for the suggested price, to ensure that the rental fee for e-bikes remains competitive with other commonly used modes of transportation (American Automobile Association, 2023; San Francisco Municipal Transportation Agency, n.d.).¹

Additionally, to create a more practical solution, we evaluated two constraints. One is the geographical constraint mentioned above. Another significant constraint is the budget limit. Perhaps it would be helpful to focus on providing subsidies to low-income groups at first to meet this constraint. These aren't obstacles that need to be overcome but rather fixed requirements to the solution that must be met.²

Methods

We sampled 35 individuals with convenience sampling due to budget and time constraints. This is extremely limited because it relies on the availability and willingness of the participants which may result in the over or underrepresentation of various groups leading to biased results.

¹ **#gapanalysis:** The gap analysis is applied through a two-step process: listing all existing solutions and evaluating them. The existing solutions are identified in detail and the reason for identification solution is suggested (two way approach to identify the current solution in the US and applicable solution that is implemented in another country). The existing solutions are used to devise a SF-specific solution. We also considered other modes of transportations as alternative solution, and suggested appropriate bike renting price based on their price, so that our suggested plan actually works effectively.

² **#constraints:** The two constraints (geographical and budget constraint) are identified along with proper differentiation between constraint and obstacle. It also discusses how these obstacles can be overcome with an adjustments in the solution.

We tried to mitigate this by randomly selecting two districts out of 12 in San Francisco using an online generator (Sumida, 2022). Randomly selected two districts were District 2, which includes Presidio, Pacific Heights, Presidio Heights, and the Marina, and District 7, which includes the Inner Sunset, West Portal, Forest Hill, and Parkmerced. We then asked individuals on the street around Trader Joe's to participate in our survey. We had a sample of 15 San Francisco residents for each district, and we manually removed 5 responses from non-residents in our results. This occurred because of the limitation of our survey design that allowed everyone to complete the survey.³

In order to develop the survey questions, we pretested with our draft questionnaire on a convenient sample of 20 students from Minerva University and made adjustments to questions as needed to remove ambiguities. We followed research principles by including a consent form, avoiding emotionally charged questions, collecting anonymous data, and making the condition that all respondents are able to withdraw their answers promptly apparent in the consent form.

We followed best practices for survey questions by giving clear instructions, keeping the question structure simple, asking one question at a time, and providing a list of acceptable responses to closed questions while providing an “other” section where participants can elaborate on their own answers (See Appendix C for survey questions). We administered the survey online using Google Forms and respondents were asked to scan a QR code if they agreed to participate. We chose to administer the survey online as it gives the respondents the ability to leave and fill out the survey on the go.⁴

³ **#sampling**: We mentioned why we opted to use convenience sampling and explained the limitations of the method. We described how we tried to mitigate the limitation by randomly selecting the districts but ultimately recognized that any results derived would be extremely limited.

⁴ **#interviewsurvey**: We aimed to gather reliable data from our surveys and followed research principles and best practices when designing the survey questions. The limitation of the study is thoroughly explained and we recognize a limitation of the survey design as well.

Demographics of survey respondents

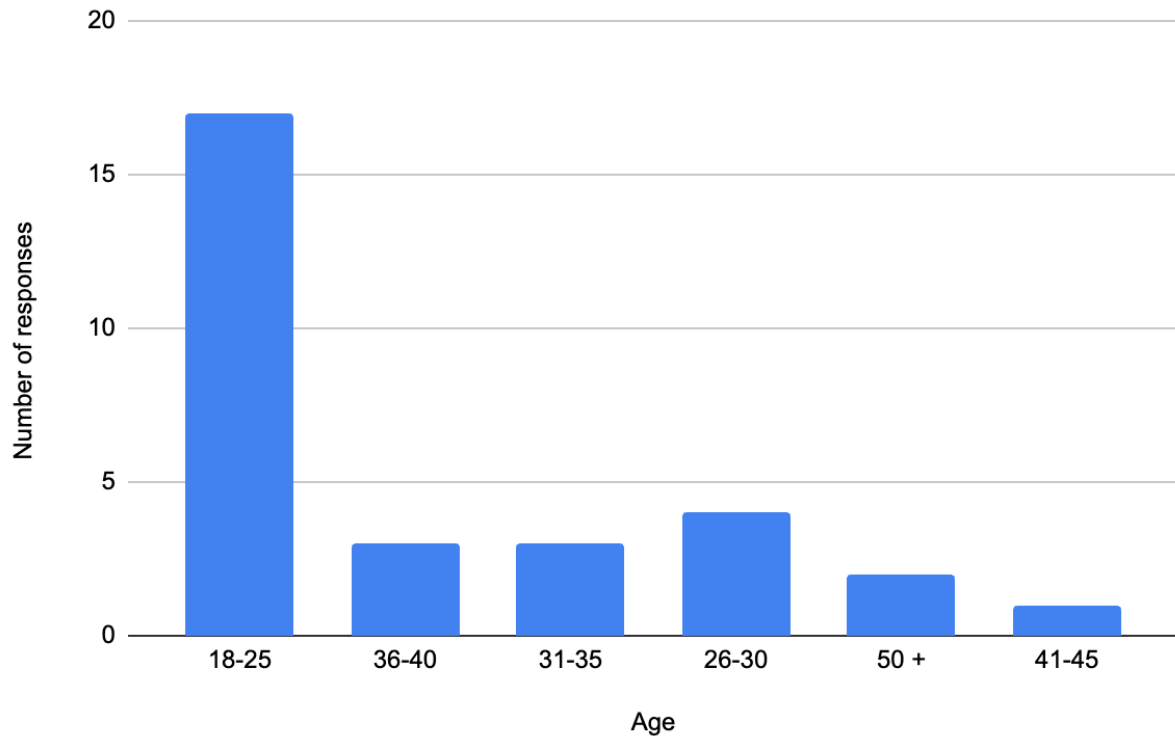


Figure 1: The figure shows the age distribution of respondents. The majority, 17 of 30 respondents are from ages between 18 and 25, making the distribution extremely skewed to the right.

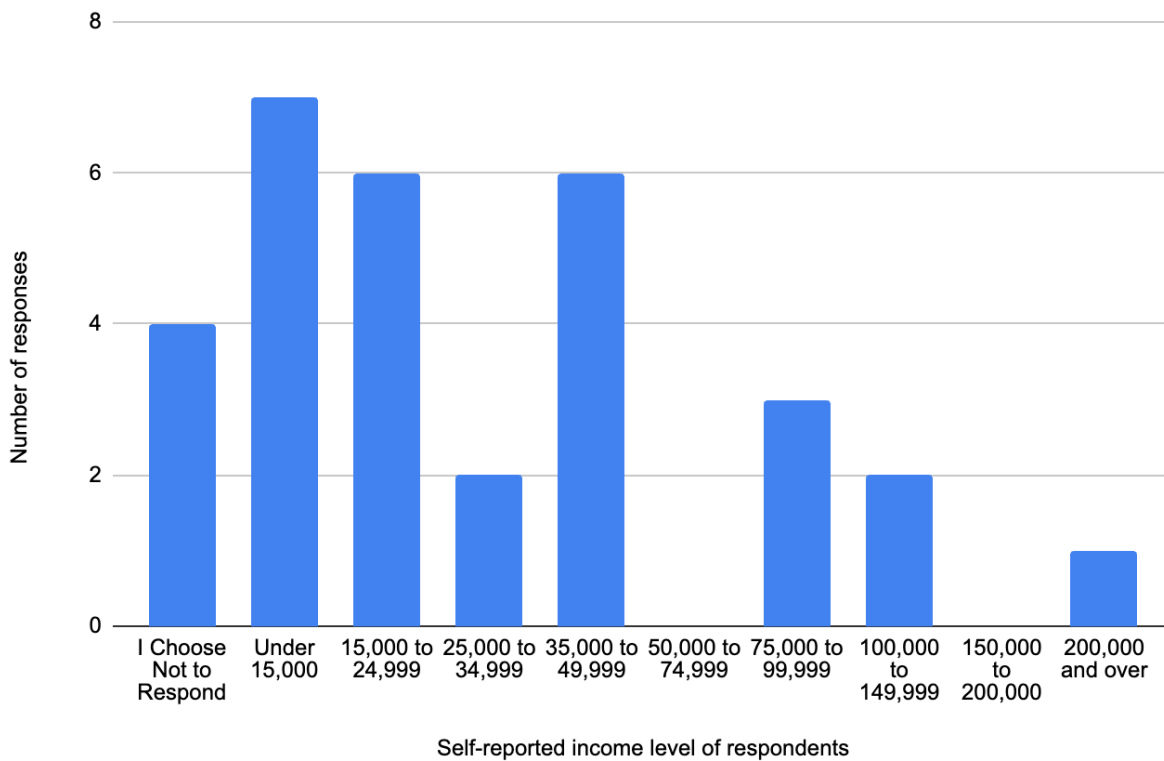


Figure 2: The figure shows the answers to the question “Select your annual household income (\$).” The majority of respondents have a self-reported annual household income of under \$50,000 and the distribution is slightly skewed to the right. 4 people have chosen not to respond.

Limitations

It is imperative to note that the small sample size of 30, the convenience sampling method, and the heavily skewed age distribution are factors that limit the representativeness of this sample. Therefore, the results drawn from this study are extremely limited.

Results

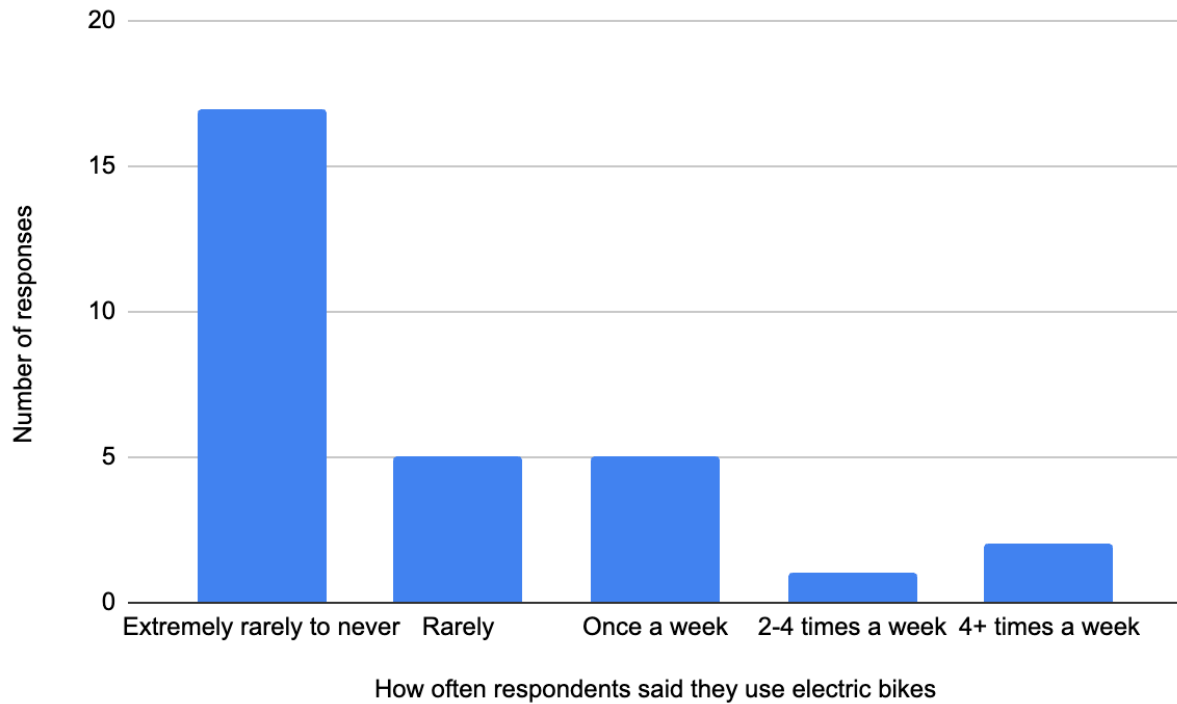


Figure 3: The figure shows the frequency of how respondents answered the question “How often do you use electric bikes as a means of transportation?” The majority, 17 of 30 respondents, answered “Extremely rarely to never.”⁵

⁵ #dataviz: We generated histograms to show the distribution of the different answers. We specified what question the answers were referring to in the captions. We made sure the graphs had axis titles and added explanations in the caption.

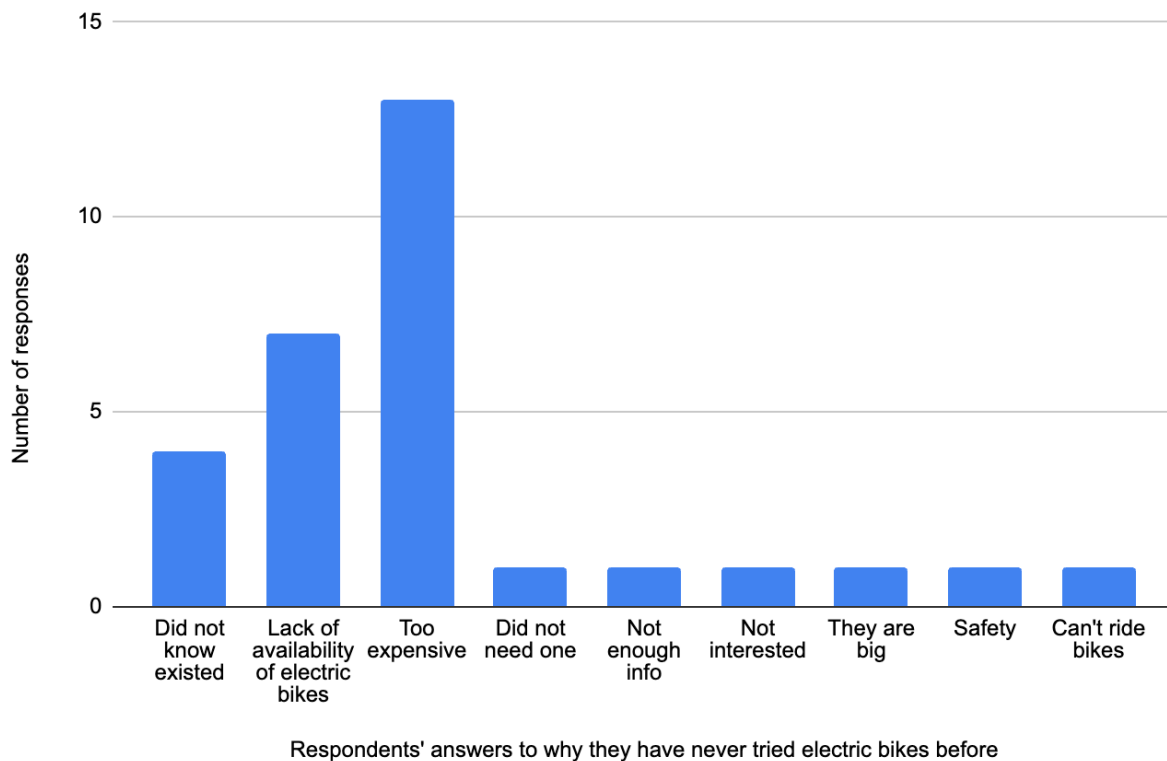


Figure 4: The figure shows the frequency of how respondents answered the question “if you have never tried electric bikes, why?” Respondents checked all those applied and were able to formulate their own response as “other.” Out of 26 unique responses, half the responses checked “too expensive.”

Summary for Mayor London Breed

The city of San Francisco has been a leading example in reducing carbon emissions even as the city population increased. However, there is still a lot to be done to reach the ultimate goal: net zero carbon emissions in 2040 (*SF Environment*, 2021, p.13). Our research has been

focused on a sector that is still leading today in San Francisco's carbon emission since 2000—local transportation. San Francisco Municipal Transportation Agency's Battery Electric Program has been a success and has reduced munis emissions to 2% of the City's transportation-related emissions (SFMTA, 2023). However, private transport (cars and trucks) accounts for 96%, equivalent to 1.2 million tons of carbon dioxide emissions as of 2020 (City and County of San Francisco, 2020).

Therefore, to reach the ultimate goal of net zero carbon emission by 2040, we decided to start by focusing our research on the subgoal that will bring us closer to our main goal. Our research has been in finding a way to maximize the efficiency of one of the city's supporting actions from the 2021 Climate Action Plan—"build a fast and reliable transit system that will be everyone's preferred way to get around." Therefore, we focused on creating subsidies for renting electronic bikes. According to SFMTA, having a single person switch and commute by bike rather than the ways mentioned above can reduce up to 1.9 tons of CO₂ emissions annually. However, regular bikes require significant physical strength and time. Considering the geography of the city of San Francisco, we prioritized electric bikes over bicycles. Affiliation with biking companies such as Lyft has increased biking to ~172,807 rides in San Francisco in the last month, with the majority being electric bikes—110,920 (Lyft, 2023). However, to reach the main goal, bike usage should increase further, so we suggested an initial solution of subsidizing the e-bike renting fee.

After creating an initial solution to promote e-bike usage, we additionally conducted a survey in order to diagnose how actual SF residents think about e-bikes. Then we analyzed the data to suggest further improvements by addressing sub-goals to reach the ultimate goal state. In the survey, three problems were most prominent in relation to the low usage of bikes and

especially electric bikes: “lack of availability,” “did not know it existed,” and “too expensive.” The response of “did not know it existed” suggested that people aren’t aware that there are electric bikes that are faster, require less physical strength, and could be parked easily. This can be addressed by increasing e-bike awareness by creating advertisements that emphasize the benefit of e-bikes. The lack of availability implies that there are not enough numbers of e-bikes and e-bike stations, so some of them were not able to access e-bikes conveniently. This problem can be addressed with the subgoal of increasing the number of e-bikes around the city. The third problem—expensiveness—was the most prominent. 54.2% of people proclaimed e-bike prices being “too expensive” (see Figure 4).⁶ We then consulted Lyft’s e-bikes currently priced at \$12.49 for a 30-minute ride. According to the US’s e-bike regulations, e-bikes should cease to assist when the eBike reaches 28 miles per hour (QuietKat, 2023). Therefore in a 30-minute ride, an e-bike could only get to 14 miles for the above price while a car could go to double that distance on one gallon of gas priced at \$3.606. Thus, after careful consideration of all factors that go into the pricing of e-bikes like the cost of their operations and maintenance, we decided to suggest the ultimate subgoals as subsidizing renting e-bikes from Lyft’s Baywheels bike stations (lowering prices to a range of \$2.5 to \$3.5 and have the city of San Francisco pay for the supplement), along with the two new subgoals to reach the final goal state of net zero carbon emission.⁷

⁶ **#interviewsurvey:** We surveyed 35 people, asking them relevant questions to our research, and after doing an analysis, we noticed three popular themes: “too expensive,” “didn’t know existed,” and “lack of availability,” to which we then suggested possible causes (relying on prior research) and came up with novel solutions to address them.

⁷ **#heuristics:** We used the means-ends heuristic to devise a plan to work towards the ultimate 2040 goal of achieving net zero carbon emissions. The plan included dividing the above ultimate goal into small doable goals. We chose three subgoals to focus on that would take us to the subgoal and, in the end, the ultimate goal (see Figure 5). We used the means-ends heuristic because the net zero goal requires many factors working together to achieve it, and there couldn’t be just one solution that would take us directly to it.

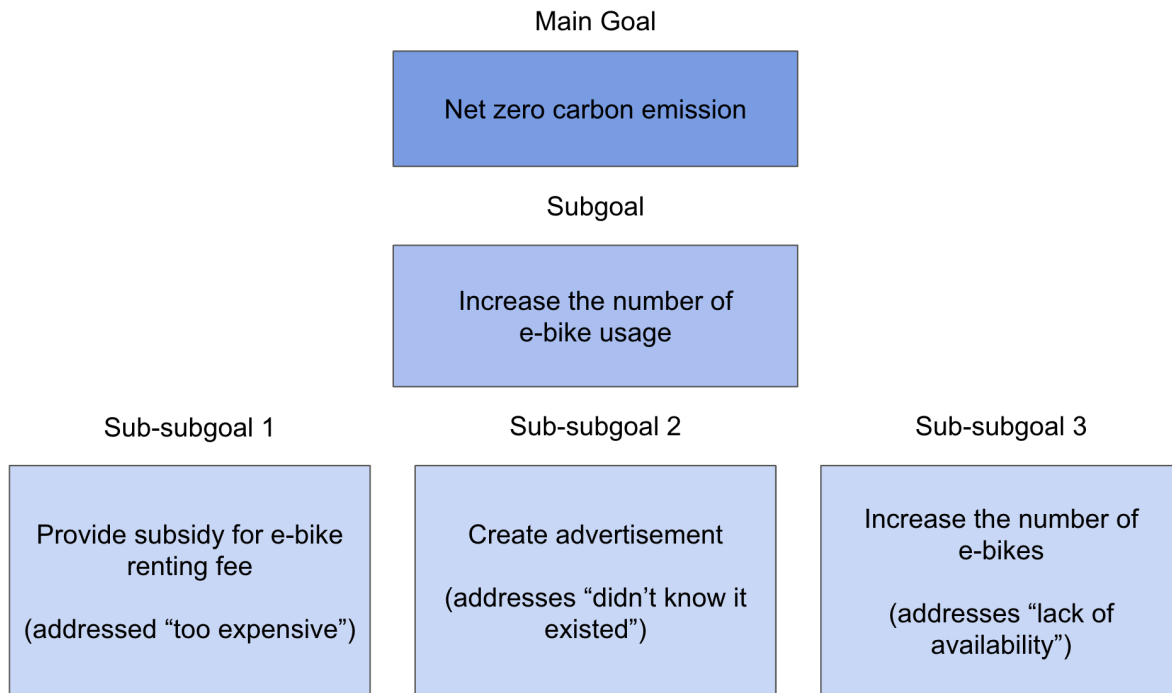


Figure 5: The figure shows the summary of the goals.

Note. We acknowledge that the subgoal (increase the no. of e-bike usage) is not the “only” way to achieve the main goal, but we intentionally did not include other subgoals (ex. other Supporting Actions) to focus on our chosen Supporting Action.

Word count: 1496 (excluding figure captions: 210)

AI usage⁸

AI was used to check grammar at an early stage of the assignment.

⁸ **#professionalism:** We followed APA conventions, proofread our assignment, and included a word count, AI usage, references, and an appendix section. Our word count abided by the 1400-1600 word limit.

Appendix

Appendix A: Consent form

Please read this consent agreement carefully before agreeing to participate in this study. *

Purpose of the research: This survey aims to investigate the utilization patterns of electric bicycles (e-bikes) among San Francisco residents, with the objective of providing recommendations for promoting e-bike usage in order to mitigate carbon emissions. To this end, this survey has been designed to elicit valuable insights from participants regarding their experiences with e-bikes, particularly regarding bike sharing service.

What you will do in this research: The survey encompasses an inquiry into respondents' means of transportation and their utilization of e-bikes in multiple-choice form.

Time required:

Participation will take approximately 3 minutes to complete.

Risks:

There are no anticipated risks associated with participating in this study.

Benefits:

By participating in this study you have helped with the education of students learning about survey research design.

Confidentiality:

Your participation in this study will remain confidential, and your identity will not be recorded. Your responses will not be connected with your name. We may associate the data with certain demographic information (such as age, income, country of origin), but that will not be associated with your name or any other identifying information (such as a Social Security number).

Participation and withdrawal:

Your participation in this study is completely voluntary, and you may withdraw at any time without penalty. You may withdraw by informing the investigator that you no longer wish to participate (no questions will be asked).

Contact:

If you have questions about this research, please contact [cmoy@minerva.edu].

Agreement:

The nature and purpose of this research have been sufficiently explained.

I understand that I am free to withdraw at any time without incurring any penalty.

If you are 18 years of age or older, understand the statements above, and freely consent to participate in the study, click on the "I Agree" button to begin the survey. If not, thank you for your time.

☐ I agree

Next

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Clear form

Appendix B: Demographics questions

Demographic information

Are you San Francisco resident? (Do you live in San Francisco?) *

☐ Yes

☐ No

Select your race/ethnicity: *

☐ Hispanic or Latino

☐ American Indian or Alaskan Native (Not Hispanic or Latino)

☐ Asian (Not Hispanic or Latino)

☐ Black or African American (Not Hispanic or Latino)

☐ Native Hawaiian or Pacific Islander (Not Hispanic or Latino)

☐ White or Caucasian (Not Hispanic or Latino)

☐ Two or More Races (Not Hispanic or Latino)

☐ I Choose Not to Respond

☐ Other: _____

Select your age *

- ☐ 18-25
- ☐ 26-30
- ☐ 31-35
- ☐ 36-40
- ☐ 41-45
- ☐ 46-50
- ☐ 50 +
- ☐ I Choose Not to Respond

Select your annual household income (\$) *

- ☐ Under 15,000
- ☐ 15,000 to 24,999
- ☐ 25,000 to 34,999
- ☐ 35,000 to 49,999
- ☐ 50,000 to 74,999
- ☐ 75,000 to 99,999
- ☐ 100,000 to 149,999
- ☐ 150,000 and over
- ☐ I Choose Not to Respond

General question about transportation

Which transportation method do you use most frequently? *

- ☐ Car
- ☐ Bicycles
- ☐ Electric bikes
- ☐ Bart
- ☐ Muni (bus)
- ☐ Walking
- ☐ Other: _____

Reason for your choice (select up to 2) *

- ☐ Convenience
- ☐ Low prices
- ☐ Time efficiency
- ☐ Environmental concern
- ☐ Privacy
- ☐ Other: _____

Which transportation method do you use least frequently? *

- ☐ Car
- ☐ Bicycles
- ☐ Electric bikes
- ☐ Bart
- ☐ Muni (bus)
- ☐ Walking
- ☐ Other: _____

Reason for your choice (select up to 3) *

- ☐ Cost
- ☐ Environmental concern
- ☐ Inconvenient
- ☐ Privacy
- ☐ Time efficiency
- ☐ Didn't know existed
- ☐ Lack of availability
- ☐ Other: _____

Questions specific to biking

Do you bike frequently? (3+ times a week) *

☐ Yes

☐ No

If no, check the top 3 reasons why

☐ Lack of infrastructure for biking

☐ Safety concerns

☐ Cost

☐ Weather

☐ Hills in San Francisco

☐ Time-consuming

☐ Lack of availability of bikes

☐ Other: _____

How often do you use **electric** bikes as a means of transportation? *

- ☐ Extremely rarely to never
- ☐ Rarely
- ☐ Once a week
- ☐ 2-4 times a week
- ☐ 4+ times a week

If you have never tried electric bikes, why?

- ☐ Did not know existed
- ☐ Lack of availability of electric bikes
- ☐ Too expensive
- ☐ Other: _____

For a 30-minute e-bike ride (currently priced at \$12.49 at Bay Wheels), how would * your electric bike usage change if the price was lowered to \$2.5 - \$3.5?

- ☐ Significantly increase
- ☐ Somewhat increase
- ☐ No change
- ☐ Somewhat decrease
- ☐ Significantly decrease

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